CARLOS ROMERO MADRID

COMPUTATIONAL PHYSICIST | HPC SCIENTIFIC SOFTWARE ENGINEER | CREATIVE TECHNOLOGIST

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github.com/carrommad · October 2026

To: Dr. Fernando Cucchietti & Selection Committee
Data Analytics and Visualization (DAV) Group · Barcelona Supercomputing Center (BSC-CNS)

STRATEGIC APPLICATION
Data Visualization, Creative Computing & Science Democratization

✦ LIVE DIGITAL DOSSIER & COMPANION WEB EXPERIENCE

INTERACTIVE

carrommad.github.io/cool-cv ↗

Explore online: Interactive European Trajectory Map · Real-Time MN5 Audio Synthesizer (MNusic) · 3D Bio-Kinetic Sensor Visualizer

Dear Dr. Fernando Cucchietti and Members of the Data Analytics & Visualization Group,

Most applications probably start with a catalog of keywords. Let me start with a sensory contrast: the deep, constant hum of **MareNostrum 5** solving climate turbulence across thousands of cores, and the crystalline chime of an ESP32-powered kinetic sphere rolling across the floor as my cat **Pipa** nudges it into motion.

I am a **Computational Physicist** (Double Degree in Physics & Mathematics from Seville, 1st of Class MSc in Plasma Physics from Aix-Marseille) and **HPC Scientific Software Engineer** in BSC's Performance Team. Daily, I deploy, couple, and optimize petascale simulation workloads across Europe's tier-0 infrastructure (MN5, Leonardo, LUMI). Yet my fundamental conviction is simple:

> *"If petabytes of scientific breakthroughs stay locked behind cold terminal logs and impenetrable matrices, we have only completed half the mission. Like Domestic Data Streamers, I believe data is not just an analytic output—it is raw material waiting to be felt, touched, and inhabited."*

In my independent creative computing studio, I turn this conviction into tangible, mathematically disciplined realities:

01 / MNUSIC ENGINE

Supercomputing Sonification: A real-time synaesthetic engine transforming live MN5 & LUMI telemetry into generative harmonic soundscapes and reactive particle fields (Web Audio API). Visitors hear the pulse of computational science.

02 / PIPA-MUSIC-BALL

Tangible Computing: An embedded kinetic sphere (ESP32 + 6-DoF IMU) mapping feline motion vectors to microtonal audio scales, proving that sensor physicalization can be mathematically rigorous, tactile, and playful.

03 / EXPO-CURATOR

Spatial Curation: An open-source gallery staging platform (github.com/carrommad/expo-curator) automating artwork classification, catalog layout, and exhibition wall geometry for scientific dissemination spaces.

I operate with high iteration velocity. Treating Large Language Models and agentic pair-programming as fluid thinking partners allows me to transition without friction between C++/Fortran HPC kernels, microcontroller firmware, browser Web Audio synthesis, and Adobe After Effects procedural visual essays.

Having followed your workshops and tracked your flagship initiatives—from high-performance urban modeling in **GDU-HPC** to environmental resilience in **AIR** and urban mobility in **REALLOCATE**—I see a profound natural convergence with my daily work. Your five core lines (*Scientific Storytelling, Interactive Viz, Digital Twins & Industrial AI, Creative Intelligence, and Urban Science*) represent the exact intersection where my computational physics grounding, digital twin workflows (DestinationE / ClimateDT), and sensory media experiments belong.

I don't just want to attend your workshops from the audience—I want to join Fernando Cucchietti and the DAV group to co-design and engineer the most daring, unforgettable scientific data installations in the world.

Sincerely and with great enthusiasm,

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